

Skin Cancer Detection from Dermatoscopic Images using Machine Learning techniques

Aditya Purohit, Anusha Upadhyay and Swasti Sadanand

Problem Statement

#1

Most Common
Cancer Type

95%

Survival Rate
with Early Detection

↓ Access

To Specialists
in Rural Areas

Goal: Binary classification — Benign vs. Malignant — as a first-line automated screening aid

Data

10,015

Total Images

7

Lesion Classes

4:1

Benign:Malignant Ratio

256px

Input Resolution (CNN)

Lesion Class Breakdown

MEL — Melanoma	1,113	MALIGNANT
NV — Melanocytic Nevi	6,705	benign
BCC — Basal Cell Carcinoma	514	MALIGNANT
AKIEC — Actinic Keratosis	327	MALIGNANT
BKL — Benign Keratosis	1,099	benign
DF / VASC — Other	257	benign

Binary Classification Target

MALIGNANT

MEL + BCC + AKIEC
1,954 images

BENIGN

NV + BKL + DF + VASC
8,061 images

4:1 imbalance → requires special handling

Train / Val / Test = 70% / 15% / 15% (stratified)

Train: 7,010 | Val: 1,502 | Test: 1,503

Approach & Network Architecture

Multi-Stage Pipeline

1

Multiclass Pretraining

EfficientNet-B5 trained on all 7 lesion classes (15 epochs) to learn rich skin texture features

2

Binary Fine-Tuning + Progressive Unfreeze

Focal Loss binary head. At epoch 10: full backbone unfrozen with halved LR — ULMFIT strategy

3

EVA02 Vision Transformer

21.7M params, ImageNet-22k pretrained. Attention-based — orthogonal errors maximize ensemble gain

4

Weighted TTA Ensemble

5-pass test-time augmentation. Weighted avg of B3+B5+B5p+ViT by validation AUC

Key Innovations

Focal Loss

 $\alpha=0.75, \gamma=2.0$ — concentrates on hard malignant cases

MixUp + CutMix

Stochastic training augmentation — improves generalization

WeightedSampler

Rebalances every mini-batch — handles 4:1 imbalance

AMP + OneCycleLR

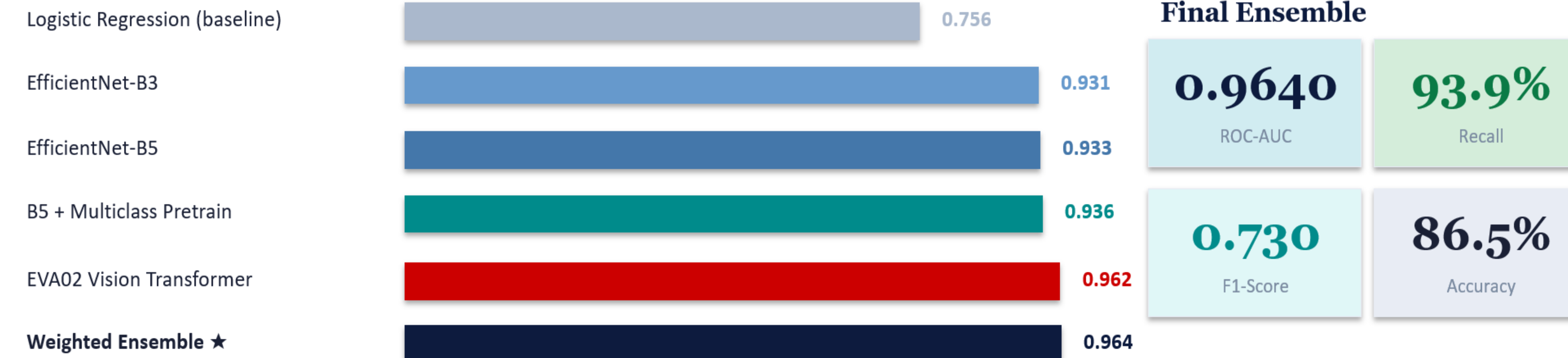
fp16 mixed precision + cosine LR schedule

Main References

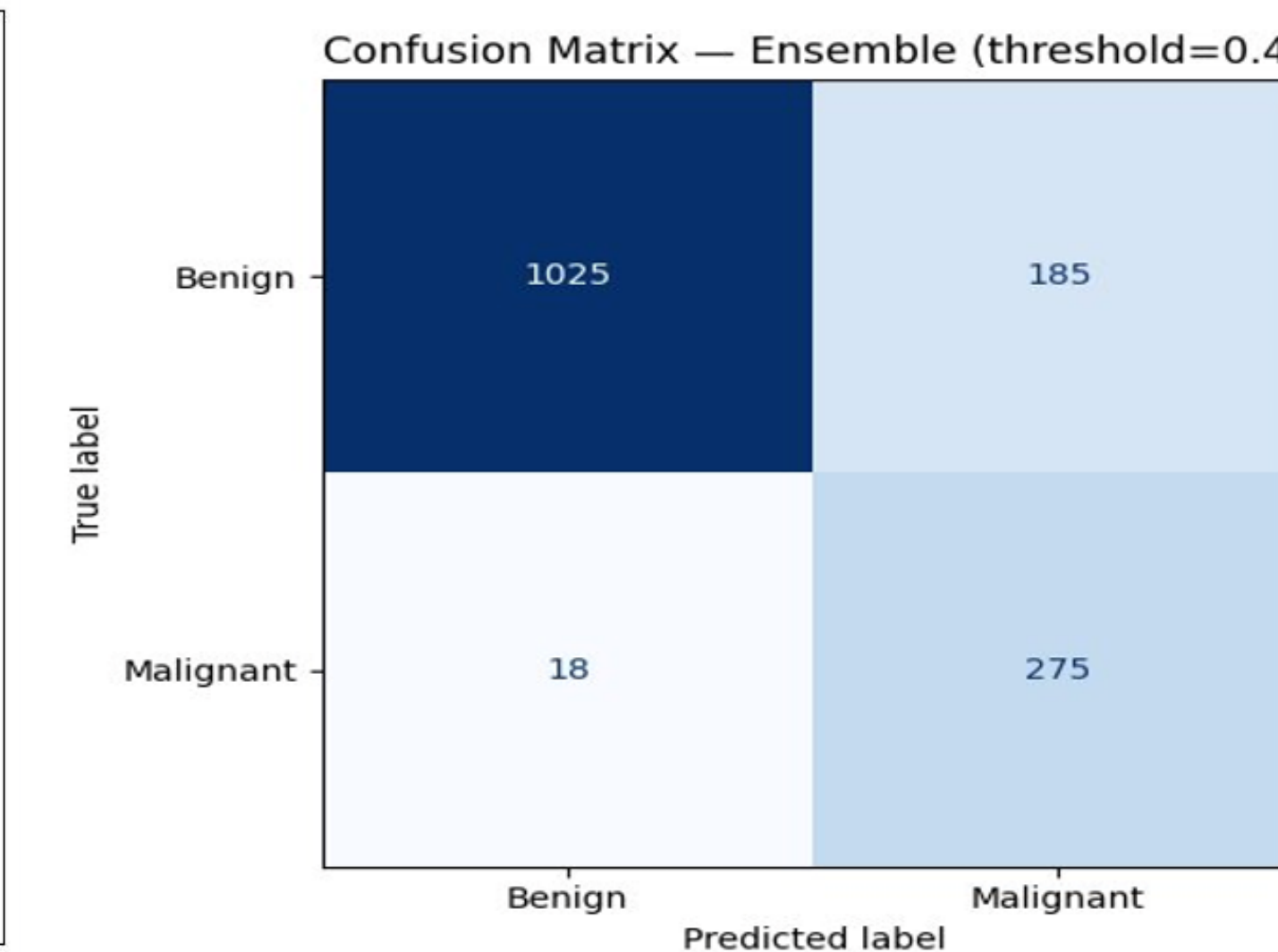
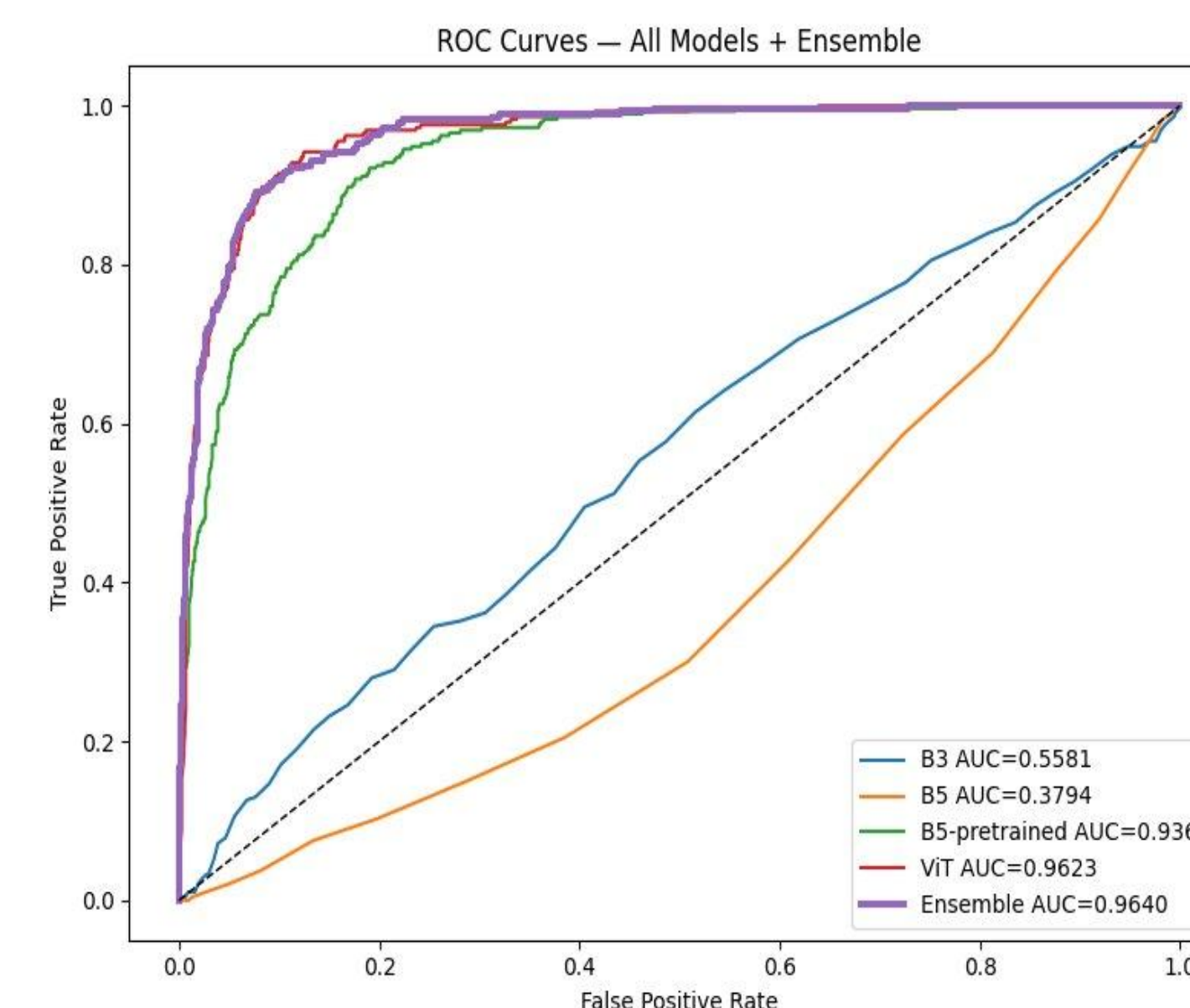
- [1] Tschandl, P., Rosendahl, C., & Kittler, H. (2018). The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. Scientific Data, 5, 180161.
- [2] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. Proceedings of the International Conference on Machine Learning (ICML).

Evaluation Plan and Results

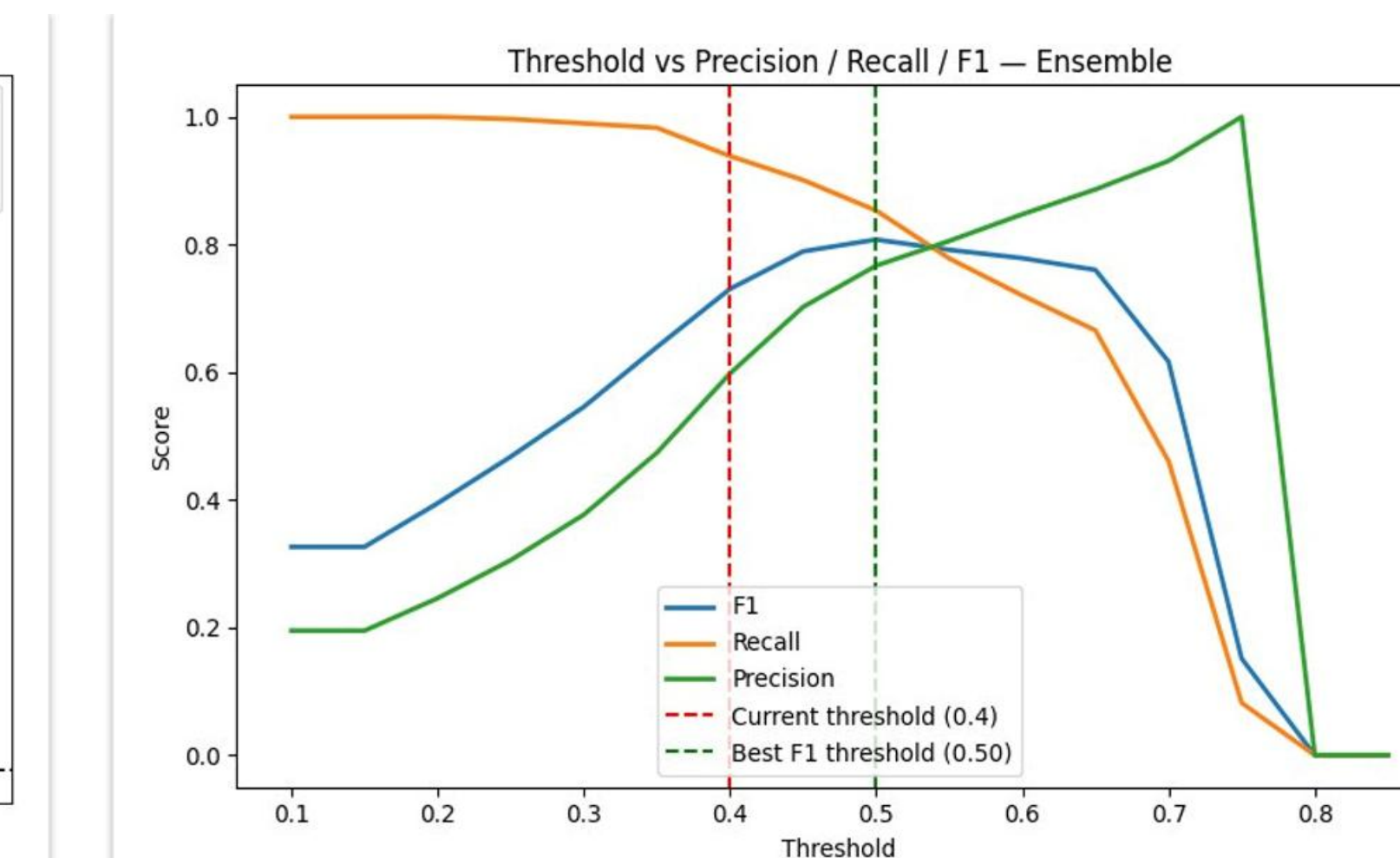
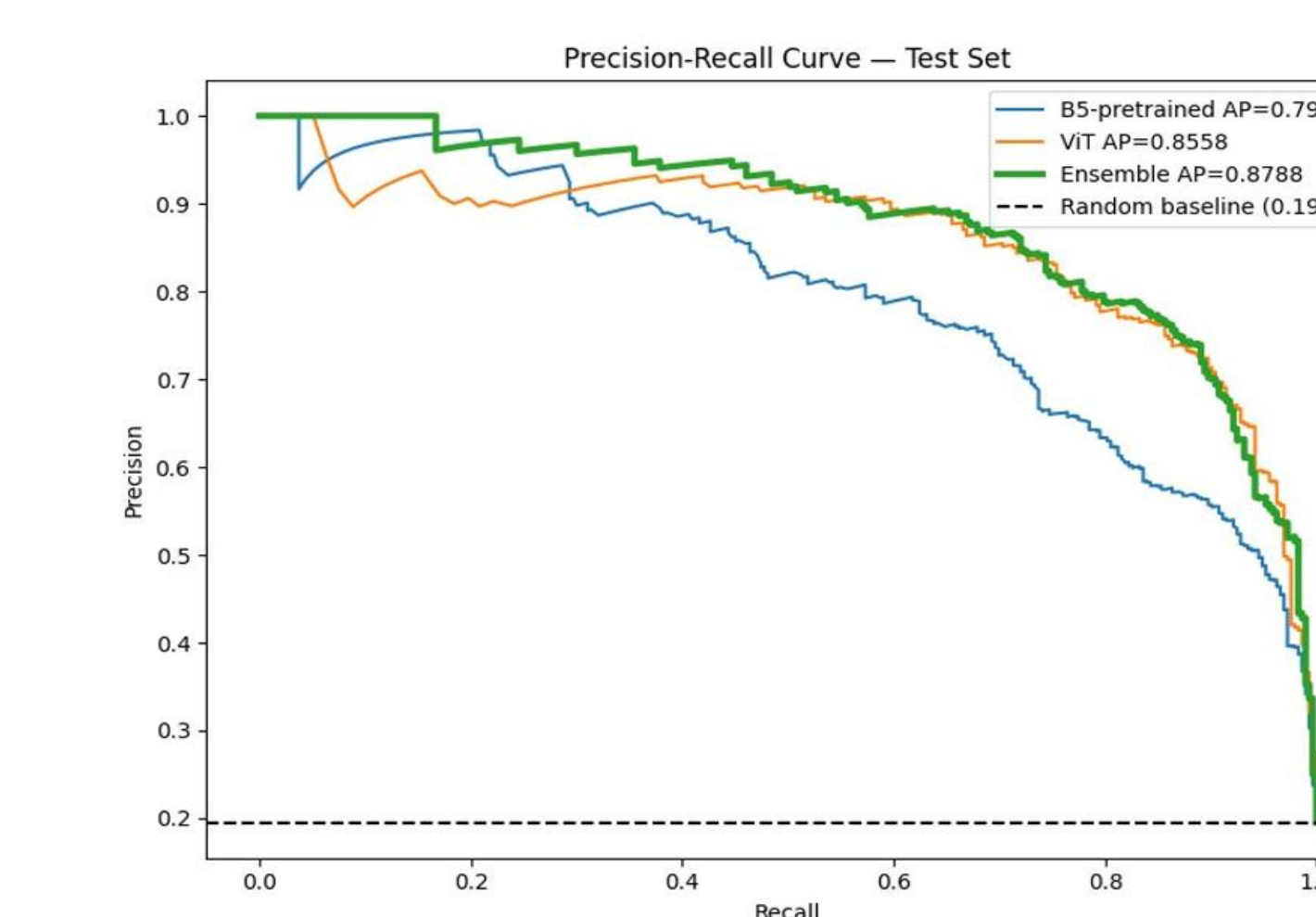
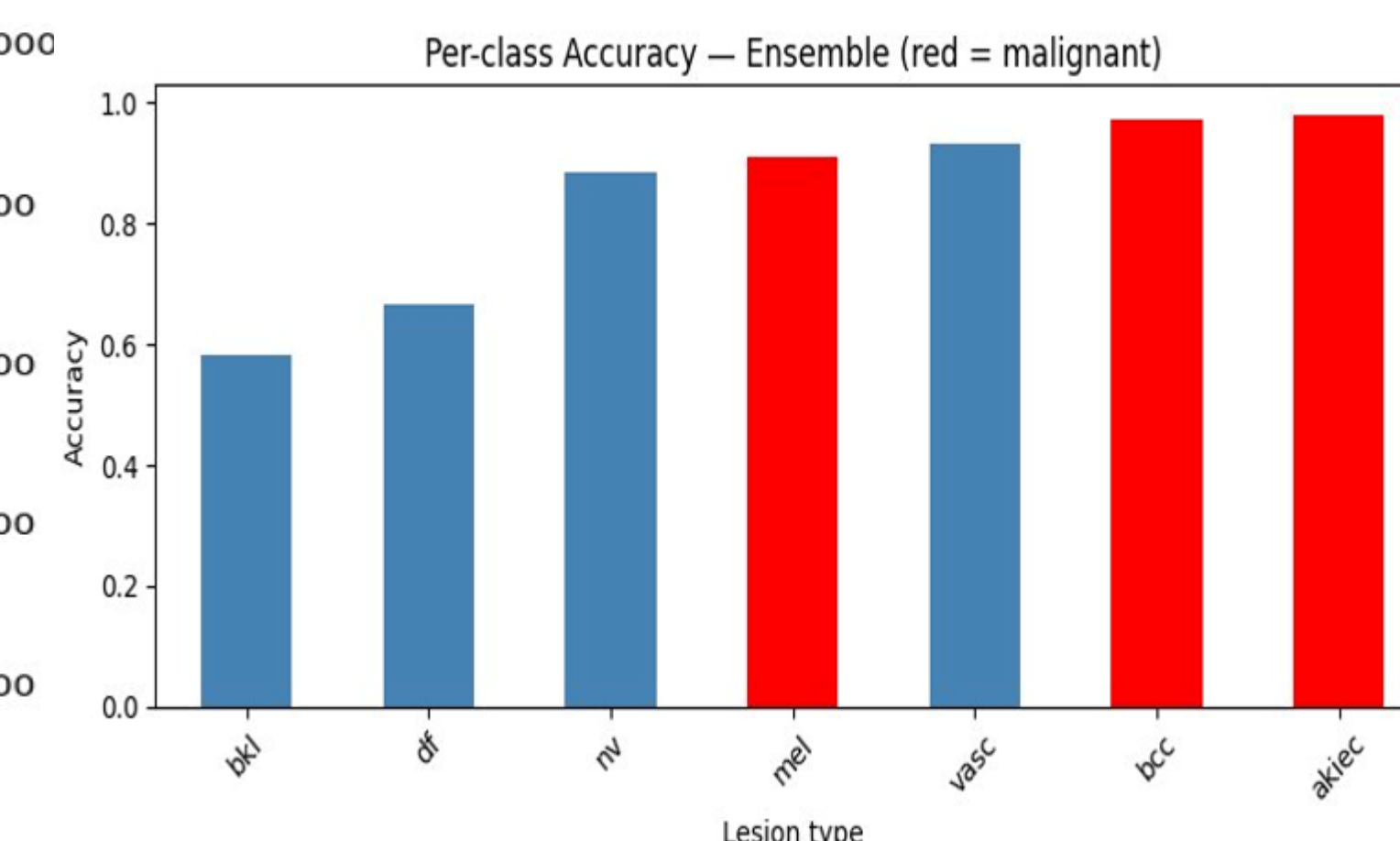
AUC Progression — Each Stage Improves



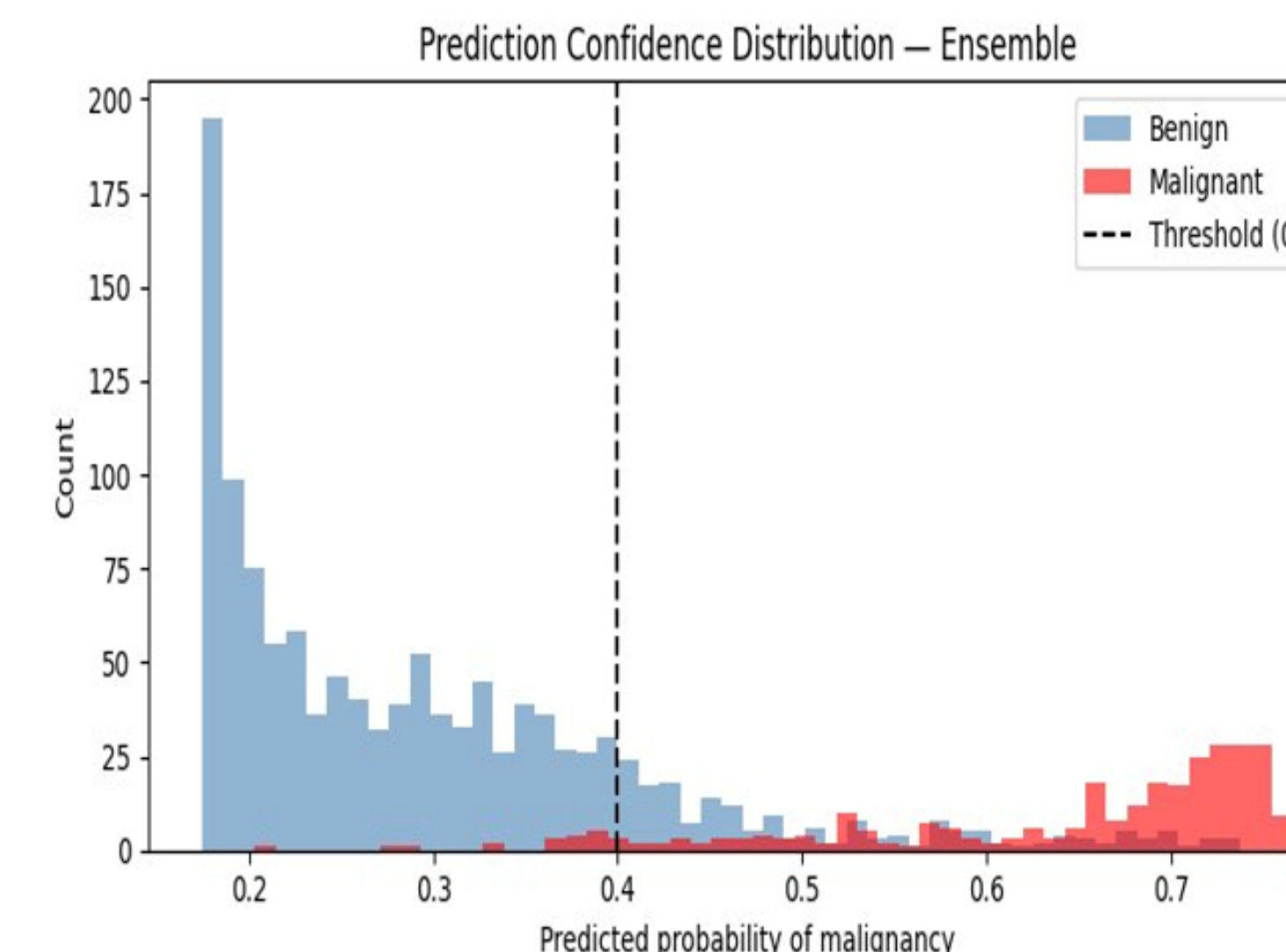
► Only 18 missed cancers out of 293 malignant cases — 94% sensitivity at clinical screening threshold



Per-Class Accuracy — Ensemble (red = malignant)



Prediction Confidence Distribution



Threshold 0.40 chosen for clinical screening: recall 0.94 > F1 0.73 — best F1 (0.808) at threshold 0.50 trades off recall

Logistic Regression	vs	Deep Ensemble
ROC-AUC	0.756 ▲ +0.208	0.964
Recall	89.1% ▲ +4.8%	93.9%
Precision	61.1% ▼ -1.1%	60.0%
F1-Score	72.5% ▲ +0.5%	73.0%
Accuracy	70.3% ▲ +16.2%	86.5%

Deep learning delivers +20.8% AUC improvement and +16.2% accuracy while maintaining high clinical recall

Key Findings

- AUC 0.964 achieved — near dermatologist-level performance (~0.94)
- Only 18 missed cancers out of 293 malignant cases in test set
- Multiclass pretraining improves binary AUC via richer feature learning
- EVA02 ViT alone: AUC 0.962, F1 0.801 — strongest single model
- Ensemble > any single model — architecturally diverse models are complementary

Future Work

ISIC Pretraining

25k+ dermoscopic images for stronger initialization

GradCAM

Visualize which lesion regions drive predictions

Uncertainty Quantification

Monte Carlo Dropout for confidence estimation

Metadata Fusion

Incorporate age, sex, anatomical site